



MINISTRY OF EDUCATION OF BELARUS REPUBLIC

Belarusian National Technical University

BNTU Branch "Scientific Research Polytechnic
Institute"

65, Nezavisimosti prospect, 220013, Minsk



TEST REPORT

23.05.2023 № 361



Scientific Research and Development Laboratory of Concrete and Construction
Materials (NIIL BiSM)

25/1, F. Skoriny street, Minsk, 220076

Phone: +375 17 272 84 18, +375 17 215 24 22

Approved by:

Head of NIIL BiSM, A.I. Bondarovich

This protocol consists of 4 pages and is executed in duplicate

Name of material (product): Complex multifunctional admixture D5 to concrete, mortar, dry building mixes, and cement, manufactured by Scientific and Production Enterprise TOKAR Co., Ltd, Russian Federation

Work performed under: Contract No. 12/23

Applicant for testing:

LLC "Arkhitekton"

Applicant's address: 9, Vachinskaya street, Putrishki agrotown, Grodnenskiy district, Grodnenskaya oblast, 231750, Republic of Belarus

Sample selection for testing was performed by: the Applicant

Sample Collection Record No. from December 28, 2022

Sample Registration Number No. 2445 from December 28, 2022

КОПИЯ ВЕРНА
дпись

1. TESTING PROGRAM

Table 1

Name of the test object (parameters, characteristics, etc.)	Test method standards	Number of samples tested
1	2	3
Water permeability		6 cylindrical specimens

Test conditions: Room temperature 19-21°C, relative humidity 90-94%.

Testing period: March 2, 2023 – March 31, 2023.

Testing was conducted at the NIIL BiSM facilities.

2. TESTING EQUIPMENT AND MEASURING INSTRUMENTS USED IN TESTS

Table 2

Designation of testing equipment and measuring instruments	Registration number	Validity period of the certificate (attestation)	Certificate number
1	2	3	4
Caliper IIIИЦ-1-150-0.01	63045251	11.12.2023	Verification Certificate No. 1-06491 36-4122
Feeler gauge set No. 4	54	13.02.2024	Verification Certificate No. 1-0124396-4123
Testo combination instrument (thermohygrometer)	62936091/106	01.08.2023	Verification Certificate No. 1-0421525-5022
Explorer (E) E0 D 110 electronic scale	1119121672	07.06.2023	BelGIM Calibration Certificate BY 01 No. 000585 8-4722
BW-30 RB electronic lab scale	020603185	07.06.2023	Calibration Certificate BY 01 No. 0005863-4722
CMЖ-435 A vibrating platform	3749	01.04.2023	Attestation Certificate No. 1261-47-A/2022
ФИ-150 cylinder moulds, 150 mm diameter	71-74	29.04.2023	Attestation Certificate No. 1974-41
ЛТ-300 electronic lab thermometer	899062	19.12.2023	Verification Certificate No. 074779-5522

3. COMPOSITION AND CHARACTERISTICS OF MATERIALS USED DURING TESTING

Table 3

Name of the mix design	Consumption of components, kg per 1 m³ of concrete mix				
	cement	sand	crushed stone	water	admixture
Main (with admixture) OK (slump) = P1	350	760	140	140	10,5 (3,0% of cement weight*)

*- dosage of the admixture in dry form

Material specifications for concrete mixes:

Binder: Portland cement CEM I 42.5 N (GOST 31108-2020), JSC "Krasnoselskstroyaterialy"

Fine aggregate: Sand (fineness modulus 2.4), GOST 8736-2014, JSC "NerudProm"

Coarse aggregate: Crushed granite (fraction 5-20 mm) (GOST 8267-93), RUPP "Granit"

Mixing water: Tap water (STB 1114-98) and potable water (STB 1188-99)

Complex chemical admixture D-5, manufactured by SPE TOKAP Co., Ltd (Russian Federation) — provided by the customer.

КОПИЯ ВЕРНА

ПОДПИСЬ



4. SPECIMEN PREPARATION AND TESTING METHODOLOGY

Water permeability

The objective of this test was to determine the "self-healing" effect of cracks in concrete specimens prepared using the complex multifunctional admixture D-5 (designed for concrete, mortar, dry mixes, and cement).

The preparation of test specimens was carried out in stages.

In the first stage, cylindrical specimens with a height of 150 mm were fabricated using the main mix design specified in Table 3.

In the second stage, after 24 hours of specimen curing, cracks were artificially induced in the cylinders by applying a load parallel to the concrete placement layers. The load was controlled to achieve a crack width of approximately 0.1-0.2 mm at the bottom of the specimens. Subsequently, the cracks formed on the sides of the specimens were sealed (bonded) using an epoxy adhesive.

In the third stage, a 2-3 cm high rim was formed around the top of the specimens and subsequently filled with water. The water level was maintained at a constant height. The specimens were then placed in a standard curing chamber at a temperature of $(20 \pm 2) ^\circ\text{C}$ and a relative humidity of $(95 \pm 5) \%$ to continue hardening.

Figure 1 shows a diagram of a specimen with a crack.

Трещина

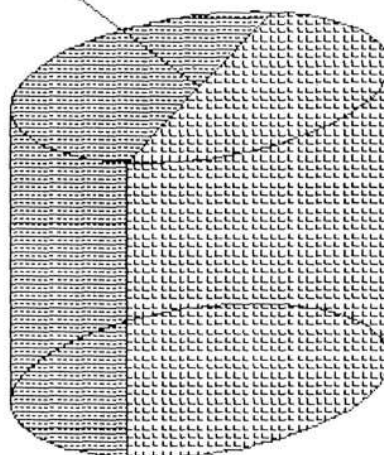


Рисунок 1 – Образец с трещиной.

5. TEST RESULTS

After one week of specimen curing, water flowed freely through the crack.

After 1.5-2 weeks, water was observed seeping drop by drop through the crack.

After 2-3 weeks, a wet spot was visible on the underside of the specimens (indicating that the crack had begun to partially seal).

By the fourth week of curing, the water leakage ceased completely, and the wet spot dried up.

ПОПЯ ВЕРНА
ПОДПИСЬ

6. CONCLUSION

When using the complex multifunctional admixture D-5 in concrete, mortar, dry mixes, and cement, the "self-healing" effect of cracks in the concrete specimens began to manifest four weeks after fabrication.

The test results apply strictly to the tested specimens provided by the Customer

Contract manager:

Researcher NIIL BiSM:

Научный сотрудник НИИЛ БиСМ

P.L. Fedorovich P.L.

Fedorovich

Младший научный сотрудник НИИЛ БиСМ

Г.С. Чикулаев

Протокол составил:

Инженер НИИЛ БиСМ

Т.А. Осипюк



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Date of issue to the Customer: May 23, 2022

The protocol is executed in duplicate:

-1 — NIIL BiSM;

-2 — LLC "Arkhittekton"

ПРОВЕРЕНО

Нормоконтролер

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End of test report



КОПИЯ ВЕРНА

ПОДПИСЬ

A handwritten signature in blue ink, likely belonging to the copy holder.